# Terms of Reference

## Working Group 3: Cryospheric Risk Management Solutions

## The Decade of Action for Cryospheric Sciences (2025-2034)

## Background

On August 13, 2024, the United Nations General Assembly adopted Resolution 78/321, proclaiming 2025 to 2034 as the Decade of Action for Cryospheric Sciences. The resolution aims to accelerate efforts to address the urgent challenges posed by the melting cryosphere through enhanced scientific research, monitoring, and international cooperation. The resolution highlights the need to understand the cryosphere's role in the Earth's climate system, freshwater resources, ecosystems, and human social, political, and economic activities. It also calls for raising awareness of cryospheric loss and encourages the sharing of best practices to build resilience against related risks. Member States, UN bodies, and stakeholders are invited to support these activities. UNESCO is tasked with leading implementation of the Decade, with other UN organizations and partners, using voluntary contributions to develop programmes.

During the first year of the Decade, 2025, an Ad Hoc Committee was formed by UNESCO to form the structures necessary to implement the Decade. These are summarised in a **Synthesis Document**, which outlines the vision for the Decade. As a result of this process, the committee identified five Overarching Challenges (OCs) for the Decade:

OC1. Global Cryosphere Monitoring
OC2 Actionable Cryosphere Projections
**OC3 Cryospheric Risk Management Solutions**
OC4 Adaptation to Cryosphere Loss
OC5 Amplify the Impact of Cryospheric Science

Within the Decade, Working Groups aligned with each OC serve as the *main engines of delivery*. The role of Working Group 3 is to achieve the Main Goals (MGs) of **OC3, Cryospheric Risk Management Solutions**:

MG3.1: A Global Cryosphere Risk Assessment
MG3.2: Cryosphere Loss Vulnerability Assessment
MG3.3: Operationalise Cryosphere Risk Management Solutions

The Decade is split into three triennia. Each MG is assigned a 3-year block, marked by a triennial summit. It is anticipated that appointment to each working group will be on an initial three-year basis, with membership rotating as focus shifts towards subsequent MGs.

By the end of the Decade, this Working Group will create a coherent pathway from risk identification, through vulnerability understanding, to actionable solutions – supporting evidence-based, community-centred cryosphere risk management throughout the Decade.

## Initial Objectives

The cryosphere is undergoing rapid change, leading to increasing risks to ecosystems, infrastructure, water resources, and human populations. These risks arise both from **acute cryosphere-related hazards**—such as glacier collapse, glacial lake outburst floods (GLOFs), rain-on-snow events, river-ice hazards, sea ice decline, and permafrost degradation—and from **longer-term systemic impacts**, including altered river seasonality, reduced meltwater availability, sea level rise, ocean current changes, and cascading effects on the water–energy–food nexus.

OC3 addresses the need to move beyond hazard identification toward integrated cryosphere risk management. The focus of the *first three years is on MG3.1 A Global Cryosphere Risk Assessment.* This will cover:

- Global cryosphere-specific hazards and impacts.
- Mapping of ongoing activities, responsible actors, and existing capacities related to cryosphere hazards.
- Identification of gaps, data needs, and missing standards for comparing cryospheric disasters.

Towards the end of the first three-year period (2026-2029), focus should move towards MG3.2: Cryosphere Loss Vulnerability Assessment.

The work of Working Group 3 over the first three years will culminate in a report to the first *Triennial Summit of the Decade of Action on Cryosphere Sciences*.

## Wider Mandate and Deliverables

As one of the Decade's five thematic Working Groups, the Working Group 3 is mandated to:

- Act as the primary delivery mechanism for OC3, reporting to and coordinated by the Strategic Management Committee;
- In the first three years (2026-2029), lead implementation of MG3.1 and support transition to MG1.2;
- Deliver tangible, policy-relevant outputs that can be showcased at the Triennial Cryosphere Summits;
- Coordinate with other Decade Working Groups;
- Engage with existing international, regional, and national cryosphere risk initiatives to ensure complementarity and avoid duplication.
    - integrate community-based observations, Indigenous and local knowledge into cryosphere risk assessment and management, with consent, recognition and benefit-sharing;

Throughout the decade, this working group will report biannually to the Strategic Management Committee, on progress towards the following deliverables:

1. An assessment of cryosphere-related risks, integrating hazard processes, exposure, and downstream impacts specific to cryosphere change.
2. Identification of the vulnerabilities of communities, ecosystems, infrastructure, ice-affected ocean regions, and river-basin systems, with explicit attention to the cryosphere–water–food–energy nexus.
3. Development and promotion of risk management solutions that mitigate the impacts of cryosphere-related hazards and changes, rather than attempting to eliminate hazards themselves.
4. Support for community-centred and transdisciplinary approaches, integrating physical science, social science, governance, and indigenous and local knowledge.

5. Strengthening of transboundary cooperation for early warning, information sharing, and adaptive water management in cryosphere-dependent regions.
6. Report on the progressive application of the globally recognized FAIR (Findable, Accessible, Interoperable, Reusable), CARE (Collective Benefit, Authority to Control, Responsibility, Ethics), and TRUST (Transparency, Responsibility, User Focus, Sustainability, Technology) principles within cryosphere risk, hazard and vulnerability information;

## Structure

### a) Open and inclusive membership

WG3 is open to all interested experts and stakeholders contributing to the objectives of the Decade. General membership shall remain open to all applicants meeting the criteria of the Decade, with no fixed threshold or numerical cap, and all members shall be entitled to participate in the discussions and activities of the Group.

### b) Leadership composition

The leadership of WG3 comprises:

- a Lead, appointed for an initial three-year term;
- two Associate Leads, appointed for an initial three-year term, jointly with the Lead, on the basis of complementarity of expertise, thematic coverage and geographical balance;
- a Secretariat, hosted by an institution with the mandate, capacity and networks to support the scientific coordination of the Working Group.

These four positions are nominated by the Ad Hoc Strategic Management Committee on the proposal of the UNESCO IHP Secretariat, following the open call for expressions of interest and considering scientific excellence, geographical and gender balance, interdisciplinary expertise, and diversity of background and generation.

### c) Core group

A flexible core group shall be established to support coordination and continuity. The size and composition of the core group shall be determined by the Lead, Associate Leads and Secretariat, in consultation with the broader WG3 and the UNESCO IHP Secretariat, on the basis of thematic needs and the level of expressed interest. The core group shall as a guiding principle include:

- a transdisciplinary mix of scientific, policy, practitioner and knowledge-holder profiles;
- geographical balance, with deliberate inclusion of expertise from polar regions, high mountains, snow and permafrost zones, and downstream / low-lying regions affected by cryosphere loss;
- dedicated seats for representatives of Indigenous peoples and local communities, in articulation with the Indigenous Advisory Committee of the Decade;
- dedicated seats for early-career scientists, as part of the next-generation capacity ambition of the Decade;
- expertise spanning cryosphere science, hydrology, hazards, social science, risk management and governance, together with regional representatives from cryosphere-dependent and high-risk regions, leaders of international programmes and partner organisations, and practitioners from development and financial sectors; Participation in the Working Group is voluntary and non-remunerated

### d) Roles and responsibilities

The responsibilities of the leadership positions are defined as follows:

- **The Lead** is responsible for the overall direction, scientific quality and delivery of WG3; represents the Group at the Strategic Management Committee and externally; chairs meetings; ensures alignment with the Synthesis Document, with the Decade's Overarching Challenges and Main Goals, and with the Triennial reporting cycle.
- **The two Associate Leads** share thematic, geographical and outreach responsibilities; act jointly with the Lead on substantive decisions; ensure thematic completeness across the breadth of cryosphere risk and hazard work; deputise for the Lead when required; and lead specific sub-streams of work as agreed within the Group.
- **The Secretariat** provides operational and institutional support to the Group; organises meetings and documentation; ensures interoperability with the host organisation's networks and existing programmes; liaises with the UNESCO IHP Decade Secretariat and with the Secretariats of the other Working Groups to foster cross-cutting coordination among the Working Groups, the Finance Dialogue and the Alliance for the Cryosphere.

### e) Term, rotation and continuity

Leadership and core-group positions are appointed for an initial three-year term, aligned with the three triennia of the Decade and the Triennial Cryosphere Summits. A transitional overlap of up to six months between outgoing and incoming officers shall be maintained to preserve continuity. Mid-term review by the Strategic Management Committee may lead to adjustments where required.

## Modus Operandi

- The Working Group shall meet primarily online (tentatively 2hrs each month), with a frequency sufficient to maintain momentum, complemented by periodic in-person meetings aligned with major international events where feasible and provided funding resources are made available.
- The Working Group shall hold at least one joint meeting per year with the other Decade Working Groups (OC1-OC5) to foster synergy, share progress on cross-cutting issues, align activities across Overarching Challenges, and avoid duplication of effort.
- Progress and outputs of the Working Group shall be reported biannually by the Core group to the Strategic Management Committee and showcased at the Triennial Cryosphere Summits. A Lead, two Associate Leads and a Secretariat will be nominated during the Committee's first meeting.
- Draft agendas and documents shall be circulated in advance of each meeting. Decisions of the Core Group and the Working Group shall be made by consensus.
- Members of the Management Committee shall be expected to consult and engage with their respective networks and constituencies, and to facilitate broad participation in the activities of the Working Groups. The Lead of the Working Group serves as a member of the Strategic Management Committee and maintains regular contact with the UNESCO IHP Decade Secretariat.
- In line with the Synthesis Document, WG3 shall not endorse, fund, promote or otherwise legitimise geoengineering research or deployment, recognising the scientifically uncertain and potentially harmful nature of such interventions. Activities shall integrate Indigenous and local knowledge ethically, with consent, recognition and benefit-sharing, and shall comply with UNESCO rules and procedures.
- Articulation with other initiatives. WG3 shall actively articulate its work with the International Polar Year (IPY 2032-2033), the UN Decade of Ocean Science for Sustainable Development (2021-2030), the International Decade of Sciences for Sustainable Development (2024-2033), and other relevant initiatives, to maximise synergies and avoid duplication.

# Website and further details

Proceedings of the Decade, consultation summaries and deliverables will be published on the Decades official website: www.un-cryosphere.org